

Who Benefits from Scientific Entrepreneurship Training?

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CGJ RCT panel · 6 sites · $N = 1,187$ startups · 14 outcomes · 26 moderators · GenericML

The Setup

What CGJ established

Theory-based (SDM) training works on average.

<i>Sample</i>	8 RCTs, 1,556 startups
<i>Outcomes</i>	Scientific learning (SI) Log sales
<i>Finding</i>	ATE positive Some HTE exists
<i>Who?</i>	Older ventures Female founders Less educated

What we do

Systematic portrait of *who* benefits.

<i>Sample</i>	6 RCTs, 1,187 startups
<i>Outcomes</i>	14 across 3 families: (1) Scientific learning (2) Business performance sales, productivity, TFP (3) Strategic behavior pivoting, exit beliefs
<i>Moderators</i>	26 baseline variables
<i>Method</i>	GenericML: BLP + GATES + CL
<i>Comparisons</i>	TE vs EB and TE vs C

The Data

Sample

	CGJ	Ours
RCTs	8	6
Startups	1,556	1,187
Control	350	332
Theory (TE)	602	428
Evidence (EB)	604	427

Missing: China, Tanzania

Demographics match CGJ

	CGJ	Ours	
Age (yrs)	33.2	33.6	✓
Female (%)	36	35.6	✓
Work exp. (yrs)	9.4	10.2	✓
Full-time (%)	40	38	✓
Dev. stage (1–5)	2.52	2.33	✓
Theory score (bl)	3.44	3.48	✓

Employees gap: NL/IT missing (larger firms). Not a selection issue.

Does It Work? Replicating CGJ's ATE

Spec: DiD startup FE + period FE, cluster(id). Treatment = randomized assignment (ITT).

Learning (SI) — we replicate

	CGJ	Ours
Theory score	2.08***	2.51***
Empirics score	1.70***	2.26***
Wald p (TE=EB) theory	0.002	0.619

Cumulative scale. Wald gap: 6 vs 8 sites.

Termination — we replicate

	CGJ	Ours
Dropout idea	0.05**	0.035
p (TE=EB)	0.964	0.948

Performance — attenuated but consistent

	CGJ	Ours
Log sales	0.53***	0.12
Productivity	0.62***	-0.01
TFP	0.53***	-0.00

Wald p (TE>EB):

Productivity	0.004	0.020
TFP	0.008	0.027

$\hat{\beta}$ attenuated: site composition + N .

TE>EB ordering preserved.

Act I

Who learns more from the training?

Act I — Learning: Two Different Stories

Empirics score (si_empirics)

β_2 (HTE signal) = 0.14, ns

G2 – G1 gap = 0.04, ns

Verdict: uniform.

Everyone gains equally in evidence collection, evaluation, and decision-making skills.

CLAN: G2 has less hours, shorter tenure, earlier stage — but the gap isn't significant enough to matter.

Theory score (si_theory)

β_2 (HTE signal) = **0.54**, $p=0.013$**

G2 – G1 gap = **+0.22*, $p=0.027$**

Verdict: heterogeneous.

Some founders absorb the causal framework significantly more than others.

Who absorbs more? → next slide.

Act I — Who Absorbs More Theory? (CLAN si_theory)

G2 = top 50% predicted benefit on si_theory. CLAN: mean of G2 vs G1 on baseline Z_i .

G2 (learns more theory) has:

Variable	G2–G1	p
exit_probability	▼ lower	<0.001
wait — G2 has HIGHER	▲ +11.8	
idea_aspiration	▼ lower	<0.001
idea_breadth	▼ lower	<0.001
hours_worked	▼ lower	<0.001
major_pivot_prob	▲ higher	0.012
months_working	▼ lower	<0.001

Portrait of the founder who learns more theory:

- ▲ **Higher exit probability** (+11.8)
- ▲ **Higher pivot probability** (+4.1)
- ▼ Lower aspirations (scale)
- ▼ Less committed (hours, tenure)
- ▼ Earlier stage

The open/flexible founder:

Not over-committed, seriously considering

Act II

Who converts learning into performance?

Act II — Performance HTE Is Massive

β_2 = signal of treatment effect heterogeneity. Larger = more heterogeneous.

TE vs EB

Outcome	β_1	β_2
Log sales	0.52***	0.51***
Log(sales/emp)	0.61***	0.67***
TFP	0.63***	0.62***
Employees	ns	0.55***
SI theory	ns	0.54**
SI empirics	ns	ns

TE vs Control (our addition)

Outcome	β_1	β_2
Log sales	ns	0.79***
Log(sales/emp)	ns	0.94***
TFP	ns	0.94***
Sales dummy	0.08***	0.80***
SI theory	0.15**	ns
SI empirics	0.53***	ns

SI vs C: large ATE, **no HTE** — everyone learns equally. Performance vs C: zero ATE on average, **massive HTE**. Training deposits knowledge uniformly. Who converts it is not

Act II — The Gaps Are Large (GATES)

Median split: G2 = top 50% predicted benefit. How large is the gap?

Outcome	Comparison	G1 (bottom 50%)	G2 (top 50%)	G2–G1
Log sales	TE vs EB	+0.11 ns	+0.94***	+0.83**
Log sales	TE vs C	−0.84**	+0.86**	+1.73***
Log(s/emp)	TE vs EB	−0.05 ns	+1.25***	+1.30***
Log(s/emp)	TE vs C	−0.90**	+1.23***	+2.08***
TFP	TE vs EB	−0.08 ns	+1.33***	+1.38***
TFP	TE vs C	−1.09**	+1.44***	+2.45***

TE vs Control reveals what TE vs EB hides:

G1 is not just unaffected — theory training *hurts* G1 founders on productivity. For the wrong founder, the framework misallocates attention.

The Paradox: Two Opposite Profiles

Who learns *more theory* vs Who *performs* better

G2 FOR SI_THEORY (TE vs EB)
absorbs most causal theory

exit_probability	▲ higher (+11.8)
major_pivot_prob	▲ higher (+4.1)
aspirations	▼ lower
hours worked	▼ lower
tenure	▼ shorter
experience	— ns

Open, flexible, not yet anchored.

≠

G2 FOR PERFORMANCE (TE vs C)
converts training into sales/TFP

exit_probability	▼ lower
major_pivot_prob	▼ lower (−11.0)
aspirations (scale)	▼ lower
hours worked	▼ lower
experience	▲ higher (+4 yrs)
age	▲ older (+3 yrs)

Experienced, committed, focused.

What this means

Who Is G2? The High-Performance Founder

Moderator	G2–G1	Sig/26
<i>Human capital:</i>		
Work experience	▲ +4 yrs	17/26
Managerial exp	▲ +1 yr	19/26
Age	▲ +3 yrs	21/26
<i>Strategic posture:</i>		
Scale aspiration	▼ lower	16/26
Pivot probability	▼ lower	18/26
Hours worked	▼ lower	20/26
Idea breadth	▼ narrower	13/26
Startup phase	▼ earlier	21/26

The theory:

G2 arrives with accumulated human capital (experience, management background, age) but has *not* over-committed: focused idea, believes in it, not betting everything on scale.

The causal framework gives them the **missing structure**: a way to systematically test what experience already suggests.

CGJ found: months working+, female+, less educated+.

We confirm experience/age.

New: aspiration and commitment are the strongest discriminators in the CGJ data.

Robustness: OLS Confirms Direction

Simple parametric check: $Y_{it} = \alpha_i + \alpha_t + \beta_{TE}(D_{TE} \cdot \text{Post}) + \beta_{INT}(D_{TE} \cdot \text{Post} \cdot Z_i) + \dots$
 Z_i standardized (mean 0, SD 1). Startup FE + period FE, cluster(id).

Outcome	Moderator	$\hat{\beta}_{INT}$	p	CLAN prediction
TFP	major_pivot_prob	-0.374	0.034**	G2 lower ✓
Log(s/emp)	idea_aspiration_2	-0.466	0.072*	G2 lower ✓
TFP	idea_aspiration_2	-0.489	0.082*	G2 lower ✓
Log sales	idea_aspiration_2	-0.382	0.089*	G2 lower ✓
Log(s/emp)	major_pivot_prob	-0.254	0.105	G2 lower ✓
Any	work_experience	mixed	ns	—

Signs consistent with GenericML CLAN in all significant cases.

OLS single-moderator is less powered than GenericML joint — expected.

Experience not significant in OLS (absorbed by correlates in multivariate GenericML).

Summary

1. Training works on learning — for everyone.

SI ATE positive and uniform vs Control. Empirics: no HTE. Theory: some HTE ($\beta_2=0.54^{**}$) — open/flexible founders absorb more.

2. Performance conversion is massively heterogeneous.

$\beta_2=0.51-0.94^{***}$ for sales, productivity, TFP. Average ATE ≈ 0 vs Control — driven entirely by heterogeneity.

3. The wrong founder is hurt.

TE vs C: G1 performance = -0.84^{**} to -1.09^{**} (sales, productivity, TFP). Theory training can misallocate attention for G1 founders.

4. Two opposite profiles.

Who learns more theory: open, flexible, high exit/pivot prob.

Who performs better: experienced, committed, focused, low exit/pivot prob.

Next Steps

Mechanism test

Does SI at P1 predict G2 membership for performance?

$T \rightarrow \Delta SI \rightarrow \text{performance}$, heterogeneous by founder type.

Does the G2 founder learn *differently*, or just convert differently?

P1 robustness

Pivoting outcomes restricted to P1 (before attrition diverges). Clean causal identification for behavioral effects.

Cognitive battery (Diego)

Risk aversion, uncertainty aversion, learning orientation.

These are the one family of moderators we haven't tested.

Could explain the G1 vs G2 conversion gap directly.

Targeting implications

If G1 is hurt, who should be enrolled?

Welfare analysis: program ROI by founder profile.

Appendix: Full BLP Table

Appendix: CLAN — Performance Outcomes

TE vs EB

Moderator	log_s	log(s/e)	TFP
major_pivot_prob	−5.9***	−7.9***	−11.8***
idea_aspiration_2	−7.3***	−7.4***	−8.3***
hours_worked	−3.3**	−7.2***	−6.3***
work_experience	+4.3***	+3.1***	ns
age	+3.2***	ns	ns
exit_probability	ns	−2.9**	−4.5***

TE vs Control

Moderator	log_s	log(s/e)	TFP
major_pivot_prob	−11.0***	−8.8***	−6.8***
idea_aspiration_2	−5.6***	−8.2***	−7.9***
idea_aspiration_1	+7.2***	+7.5***	+6.8***
exit_probability	−7.0***	−6.3***	−6.2***
idea_breadth	−5.0**	ns	−7.9***
work_experience	ns	+6.4***	ns

Appendix: Behavioral Outcomes (ATE)

Attrition: Treated startups ~ 18 pp more likely to stay in sample.
Behavioral outcomes at P2–P5 = selected subsample (most active).
Robustness: restrict to P1 (pre-attrition divergence) $\rightarrow$ pending.